

# compute\_shapley\_variance\_components.R

September 15, 2026

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| compute_shapley_variance_components                  |  |
| <i>Compute Shapley-Allocated Variance Components</i> |  |

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## Description

Converts a long-format variance-partitioning table into per-age percentages for Abiotic, Associations, and Spatial components using equal-split Shapley allocation.

## Usage

```
compute_shapley_variance_components(data_source)
```

## Arguments

|             |                                                                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data_source | A non-empty data frame with columns 'age', 'component', and 'R2_Nagelkerke'. Missing intersection labels are treated as zero, and negative RB2 values are clamped to zero. |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Details

Each two-way intersection is split equally between its two components, and the three-way intersection is split equally among all three components. The adjusted values therefore sum to the total of all seven clamped fractions.

## Value

A tibble with exactly three rows per age and columns 'age', 'component', 'R2\_Nagelkerke\_adjusted', and 'R2\_Nagelkerke\_percentage'.

## See Also

[extract\_jsdm\_variance\_fractions()], [aggregate\_jsdm\_variance\_components()]

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